Behavioral Latency as Weak Event-Time Supervision for EEG Reaction-Time Decoding

Ayana Mussabayeva[1*]
ayana.mussabayeva@mbzuai.ac.ae

Anuar Aimoldin[1*]
anuar.aimoldin@mbzuai.ac.ae

Yedige Mussabayev[2]
ymussabayev@hof-university.de

Xue Liu[1,3]
steve.liu@mbzuai.ac.ae

Kun Zhang[1,4]
kun.zhang@mbzuai.ac.ae

[1]Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, UAE
[2] Hof University, Hof, Germany
[3] McGill University, Montreal, QC, Canada
[4] Carnegie Mellon University, Pittsburgh, PA, USA
[*]Equal contribution

Keywords: weak event-time supervision, event-time posterior modeling, EEG reaction-time decoding, behavioral latency, shifted-crop diagnostics

Abstract

Single-trial EEG analyses are often organized around events and latencies, such as stimulus onset, response execution, ERP component timing, and movement onset. Yet EEG-based reaction-time prediction is commonly formulated as scalar regression from a fixed stimulus-locked window, treating reaction time as a label attached to the whole segment rather than as behavioral evidence about when a response-relevant event occurs within post-stimulus dynamics. Here we reformulate trial-wise reaction-time decoding as event-time posterior modeling. Instead of predicting a single RT directly, the model estimates a posterior distribution over response-relevant event times, $p(t_{\text{event}} \mid X)$, and reads out RT as the posterior expectation. This converts a scalar behavioral latency label into weak event-time supervision over the post-stimulus EEG window.

We evaluate this formulation on the Healthy Brain Network contrast change detection EEG task under a subject-disjoint, release-separated protocol. Architecture and readout controls show that the improvement is not explained by architecture scale or by a differentiable expectation readout alone. Distributional event-time supervision yields lower held-out error than the strongest scalar and temporal-readout baselines. Loss ablations and five-seed checks support the interpretation that the gain comes from supervising the event-time distribution, not merely from using a soft-argmax readout. EventNLL provides a probabilistic latent-event alternative, treating observed RT as a noisy realization of an unobserved response-relevant event time.

Beyond point prediction, the event-time posterior makes temporally resolved evaluation possible rather than limiting assessment to scalar nRMSE. Posterior width, target-aligned mass, mode-mean disagreement, and interval coverage characterize whether predictions are localized, diffuse, overconfident, or miscalibrated. We further introduce shifted-crop inference as a shortcut-vs-localization diagnostic: under temporal crop shifts, event-time models move in a more localizer-like direction, while also showing that fixed-window training does not fully solve shift-equivariant event localization. Together, these results support event-time distribution modeling as a probabilistic and interpretable formulation for linking single-trial EEG dynamics to behavioral timing. The same event-time view is naturally relevant to EEG settings in which the target has a latency interpretation, including ERP latency estimation, movement onset decoding, and other latency-varying neural or behavioral phenomena.

1 Introduction

Behavioral timing provides a direct link between neural dynamics and behavior. In reaction-time (RT) tasks, the observed response latency reflects sensory encoding, decision formation, motor preparation, and response execution unfolding over hundreds of milliseconds. EEG analyses often exploit this temporal structure through event alignment, response locking, and latency variability; trial-to-trial latency variation can carry behaviorally meaningful information that is obscured by averaging or by models that collapse temporal structure into window-level labels (Jung et al., 1998; Ouyang et al., 2017).

Supervised EEG decoding, however, often treats a fixed stimulus-locked window as an input to a class label or scalar behavioral variable. For RT decoding, this means that a delayed behavioral event is treated as a property of the whole EEG segment. A scalar predictor may therefore achieve low error by learning subject-level slowness, trial difficulty, or stimulus-locked response tendencies without representing when the response-relevant event occurs within the window.

We address this mismatch by treating behavioral RT as weak event-time supervision. The button press is not a direct annotation of neural event onset; it is a noisy behavioral observation produced by sensory, decision, and motor processes. Nevertheless, it constrains the timing of response-relevant dynamics. Given an EEG window X, the model estimates a posterior distribution $p(t_{\text{event}} \mid X)$ over response-relevant event times and derives the scalar RT prediction as the posterior expectation. This keeps compatibility with standard scalar metrics such as nRMSE while making the timing semantics of RT explicit.

This formulation is complementary to work on stronger EEG architectures, self-supervised pretraining, and foundation-style EEG models, which target representation learning across subjects, tasks, and recording setups (Wang et al., 2024a; Jiang et al., 2024; Chen et al., 2024; Wang et al., 2025; Ouahidi et al., 2025; Yue et al., 2025). We ask a different question: when the target is a behavioral latency, does the representation of the target itself matter, beyond backbone capacity or pretraining?

We study this question on the Healthy Brain Network contrast change detection EEG task (Shirazi et al., 2024), following the RT prediction setting introduced by the EEG Foundation Challenge (Aristimunha et al., 2025; EEG Challenge, 2025). Under a subject-disjoint, release-separated protocol, we compare direct scalar regression, temporal-readout regression, time-only segmentation, and distributionally supervised event-time models. This control structure separates architecture capacity, differentiable expectation readout, scalar timing loss with a temporal head, and full posterior supervision.

The posterior view also changes what can be evaluated from each trial. Posterior width, target-aligned mass, mode-mean disagreement, and interval coverage characterize localization, uncertainty, and calibration even when scalar errors are similar. We further introduce shifted-crop inference as a shortcut-vs-localization diagnostic: a crop-relative event localizer should shift its predicted crop-relative event time opposite to the crop shift, whereas a crop-invariant scalar shortcut should remain closer to fixed absolute timing.

Our results support this event-time formulation. Distributional event-time supervision improves held-out prediction over scalar and temporal-readout controls, remains stable in five-seed checks, and is not explained by soft-argmax alone. Posterior-geometry and shifted-crop analyses reveal temporal evidence and partial localization behavior hidden by scalar metrics. Although our experiments focus on RT prediction, the same event-time view is relevant to EEG targets with latency semantics, including ERP latency estimation, movement onset decoding, error-related responses, and other latency-varying phenomena.

Our contributions are:

1. We recast behavioral RT labels as weak event-time annotations and formulate EEG RT decoding as event-time posterior modeling rather than direct scalar regression.

2. We evaluate this formulation under a subject-disjoint, release-separated HBN-EEG

protocol against compact scalar regression, temporal-readout regression, time-only segmentation, and standardized external EEG architecture controls.

3. We show through loss ablations and five-seed robustness checks that distributional event-time supervision improves over scalar, temporal-readout, and time-only controls, and we evaluate EventNLL as a latent event-time likelihood in which observed RT is treated as a noisy realization of an unobserved response-relevant event.

4. We use posterior geometry and shifted-crop inference to separate scalar accuracy from temporal evidence, uncertainty, shortcut behavior, and partial event localization.

2 Related Work

2.1 Event-Centered EEG Analysis and Latency Variability

EEG is often analyzed through events and latencies. Stimulus onset, response execution, error-related activity, movement onset, ERP component timing, and other temporally localized phenomena define not only what occurs in a trial, but also when task-relevant neural or behavioral processes unfold. This event-centered view appears in annotation systems such as Hierarchical Event Descriptors (Bigdely-Shamlo et al., 2016) and in single-trial ERP analysis, where ERP-image visualization and related methods show that sorting trials by RT can reveal performance-linked temporal structure (Jung et al., 1998). RIDE and subsequent reviews further emphasize that intra-subject ERP latency variability can be estimated and can carry behavioral meaning (Ouyang et al., 2011, 2017).

Response- and movement-related EEG provide related examples. Lateralized readiness potentials index response preparation and execution (Coles, 1989); error-related negativities provide response-locked signatures of error commission and performance monitoring (Gehring et al., 1993; Holroyd and Coles, 2002); and movement-related cortical potentials, including the Bereitschaftspotential, organize EEG dynamics around movement onset and premovement preparation (Kornhuber and Deecke, 1965; Shibasaki and Hallett, 2006). Decision-time models make a related point at the behavioral level by treating RT variability as a consequence of latent temporal dynamics in evidence accumulation and decision formation (Ratcliff and McKoon, 2008; O'Connell et al., 2012; Kelly and O'Connell, 2013). We do not fit such a cognitive process model; we use the weaker premise that observed RT is a noisy behavioral latency associated with an unobserved response-relevant event time.

Explicit EEG event detection is also well established when annotated onsets, offsets, or intervals are available. Sleep-event detectors, for example, estimate when transient events such as spindles and K-complexes occur rather than predicting only a window-level label (Chambon et al., 2019; Tapia-Rivas et al., 2024). Our setting differs because we do not begin with manually annotated neural event times; RT is recorded as a scalar but provides weak time-of-occurrence evidence.

2.2 Fixed-Window EEG Decoding of Latency-Structured Targets

Modern supervised EEG decoding often maps a fixed stimulus-locked or task-locked EEG window to a scalar or class label. This formulation is convenient and aligns with benchmark metrics, but it can collapse temporal structure when the target itself has time-of-occurrence semantics. RT is evaluated as one scalar per trial, yet it denotes the latency of a behavioral response following stimulus onset.

EEG-based RT estimation has commonly followed this scalar-prediction form, including regression pipelines based on Riemannian geometry features (Wu et al., 2017) and deep single-trial EEG models for visual-stimulus RT prediction (Chowdhury et al., 2020). The EEG Foundation Challenge provides a standardized scalar benchmark using normalized RMSE (Aristimunha et al., 2025; EEG Challenge, 2025). We keep this scalar evaluation protocol, but ask whether the same label can be used internally as weak event-time supervision. This distinction matters because a scalar regressor may learn trial difficulty, subject-level response tendency, or stimulus-locked temporal priors without representing where the response-relevant event occurs. A temporal soft-argmax readout alone also does not resolve the issue if the model is still trained only through scalar RT error.

2.3 Distributional and Time-to-Event Output Modeling

Latency-structured EEG targets also motivate output distributions rather than only point estimates. Label distribution learning replaces a hard label with a distribution over related labels, which is useful when neighboring labels share metric structure or when ambiguity is meaningful (Geng, 2016). For RT, neighboring labels are ordered time bins, making a smoothed event-time distribution a natural supervision target.

Time-to-event modeling provides a complementary probabilistic language for ordered event times. Classical survival analysis and neural time-to-event models estimate distributions over event times rather than only scalar predictions or risk scores (Cox, 1972; Lee et al., 2018). Distributional evaluation also makes uncertainty and calibration observable through posterior width, coverage, likelihood scores, and proper scoring rules (Chapfuwa et al., 2023; Haider et al., 2020). We use this literature locally for finite EEG windows, not as a standard survival-analysis setting with long-horizon risk, censoring, or competing clinical outcomes. EventNLL instantiates this view by treating observed RT as a noisy realization of a latent response-relevant event time.

2.4 Relation to EEG Backbones, Pretraining, and Robust Decoding

A separate line of EEG decoding work addresses cross-subject, cross-session, and cross-dataset variability through stronger input representations. Classical approaches include Riemannian covariance-based decoding and alignment methods for reducing subject or session mismatch (Barachant et al., 2012; Jayaram et al., 2016; Zanini et al., 2018; He and Wu, 2020). Deep learning approaches pursue related goals through domain adaptation, moment matching, supervised EEG architectures, self-supervised pretraining, and foundation-style EEG backbones (Ganin et al., 2016; Sun and Saenko, 2016; Kostas et al., 2021; Wang et al., 2024a; Jiang et al., 2024; Chen et al., 2024; Wang et al., 2025; Ouahidi et al., 2025; Yue et al., 2025).

Our contribution is complementary to these input-representation approaches. We do not propose a new general-purpose EEG backbone, invariant representation loss, or pretraining strategy. Instead, we focus on output representation and supervision: when the target is a latency, the label is not only a scalar value to regress, but also weak evidence about when a relevant event occurred. External EEG architectures and foundation-style models trained from scratch therefore serve as controlled architecture baselines. What remains missing in prior work is a controlled EEG RT formulation that connects scalar benchmark evaluation, temporal-readout controls, distributional event-time supervision, and posterior-level diagnostics. We fill this gap by learning a posterior over response-relevant latencies and evaluating it as temporal evidence rather than only as a source of a scalar RT estimate.

3 Dataset and Evaluation Protocol

3.1 HBN-EEG CCD Reaction-Time Task

We use the reaction-time prediction task introduced by the NeurIPS EEG Foundation Challenge (EEG Challenge, 2025; Aristimunha et al., 2025), based on curated releases of the HBN-EEG dataset (Shirazi et al., 2024). Our analysis is restricted to the contrast change detection (CCD) reaction-time subset rather than the full multi-task HBN-EEG resource.

In the CCD task, participants viewed two flickering grating stimuli and responded after a contrast change made one grating perceptually dominant. The released target is the trial-wise reaction time from stimulus onset to button press. Each main experiment uses the same fixed 2 s stimulus-locked EEG window spanning 0.5–2.5 s after stimulus onset, sampled at 100 Hz from 128 EEG channels after excluding the reference channel. Thus each input has $T = 200$ samples. Because RT marks when a response occurs within this post-stimulus window, the task naturally supports an event-time interpretation rather than only a window-level scalar regression view.

3.2 Shared Evaluation Protocol

We use this task as a controlled evaluation setting for separating three possible sources of RT prediction performance: EEG backbone capacity, temporal readout parameterization, and distributional event-time supervision. The benchmark therefore fixes the data split, target support, input representation, and scalar evaluation rule before varying architectures and objectives in the subsequent model and diagnostic sections.

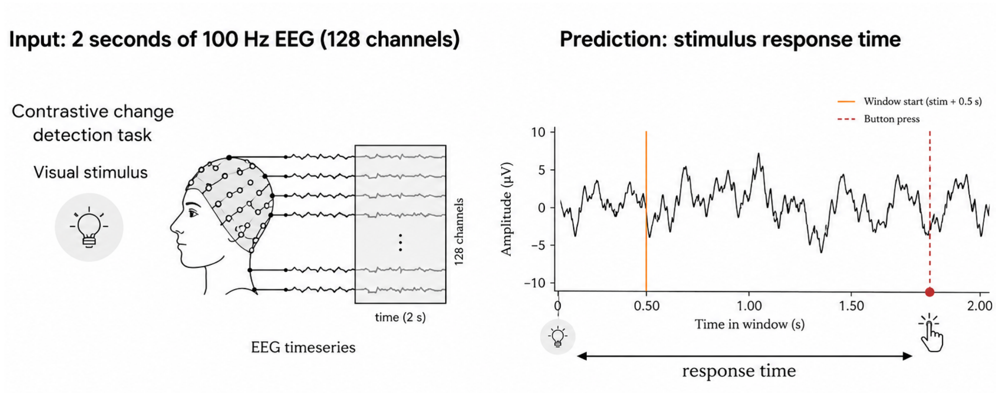

Figure 1: HBN-EEG contrast change detection reaction-time task. Each trial provides a stimulus-locked EEG segment and a trial-wise RT target from stimulus onset to button press.

Table 1: Release-separated CCD protocol statistics after window construction and RT-support filtering. Prepared trials are 2 s CCD windows with stimulus and response annotations; analyzed trials additionally satisfy $0.5 \leq \mathrm{RT} \leq 2.5$ s.

Partition	Releases	Prepared trials	Analyzed trials	Subjects
Train	R1–R8	74,576	73,030	1,221
Development	R9–R10	17,867	17,348	308
Holdout	R11	15,751	15,164	292

To avoid subject leakage, all models use the same release-separated CCD split: R1–R8 for fitting, R9–R10 for development choices such as early stopping, checkpoint selection, readout-temperature tuning, and protocol diagnostics where applicable, and R11 as a one-shot final holdout after all model and readout choices are fixed. R11 holdout labels are not used for model fitting, checkpoint selection, readout-temperature selection, stacking, or protocol decisions. For consistency, the main analyzed set includes only trials whose response lies inside the fixed 2 s window ($0.5 \leq \mathrm{RT} \leq 2.5$ s). Table 1 reports the resulting partitions; metadata checks confirmed zero subject overlap across train, development, and holdout after filtering.

Scalar performance is evaluated using normalized RMSE (nRMSE), defined as RMSE divided by the standard deviation of the true targets within the evaluated split (EEG Challenge, 2025). Main scalar comparisons use the same support-filtered R9–R10 development set and R11 holdout set. Diagnostic analyses that require shifted crop starts, stricter RT support, posterior-only summaries, or stacking are introduced locally in the corresponding subsection or appendix and do not affect the main holdout reporting protocol.

Table 2: Model-family-specific targets and readouts within the shared protocol.

Model family	Target or supervision	Readout / tuning
Direct scalar regression	Scalar RT target.	Scalar RT output; segment-pooling MLP for MSP-CNN.
Temporal-readout regression	Scalar RT target or RT-derived readout loss.	Expected time over learned time-bin weights.
External EEG backbones	Scalar RT target; supervised from scratch.	Scalar RT output.
Soft-label event-time segmentation	Gaussian soft event-time target $q(t)$.	Temperature-scaled posterior mean; τ tuned on R9–R10.
EventNLL / likelihood variants	Observed RT under a latent event-time likelihood.	Temperature-scaled posterior mean; τ tuned on R9–R10.

3.3 Shared Training, Readout Tuning, and Inference Protocol

To support this controlled comparison, we keep preprocessing, optimization, augmentation, checkpointing, and inference fixed wherever possible, so that the main experimental axes remain EEG backbone capacity, temporal readout parameterization, and distributional event-time supervision. All neural models receive the same input normalization: for each 2 s trial window and EEG channel, we subtract the channel's temporal mean within the window and divide by its temporal standard deviation within the same window. The shared training recipe uses Adam with learning rate 10^{-3}, zero weight decay, train batch size 128, up to 100 epochs, and early stopping on development nRMSE with patience 20.

Unless otherwise stated, training uses one fixed augmentation profile, so that regularization is not a model-specific tuning degree of freedom. The profile combines channel dropout, temporal cutout, and Gaussian noise with the same settings for all supervised neural models. Table 2 summarizes the remaining model-family-specific differences. Regression models use scalar RT supervision, while event-time models represent RT either as a soft crop-relative event-time target or as an observation under a latent event-time likelihood. When temperature scaling is used for event-time posterior readout, τ is selected on R9–R10 to minimize development nRMSE of the posterior-mean readout and then applied unchanged to R11. The main regression-baseline and event-time-objective comparisons are repeated over five seeds (2025–2029).

4 Regression Controls

4.1 Scalar Regression Baselines

The regression baselines establish how far scalar supervision can go before introducing event-time targets. We separate two types of controls: compact task-specific temporal controls and supervised external EEG backbones trained from scratch under the same protocol.

The compact controls form a progression of window-level temporal inductive biases.

Table 3: Direct-regression and wrapped external backbone baselines. Values are mean ± sample standard deviation.

Model	Role	Valid nRMSE ↓	Holdout nRMSE ↓
Task-specific regression baselines			
MSP-CNN	segment-pooling scalar	0.9006 ± 0.0051	0.8998 ± 0.0080
ETR-CNN	temporal readout	0.9008 ± 0.0060	0.8977 ± 0.0068
ETR-CNN large	capacity ablation	$\mathbf{0.8972 \pm 0.0040}$	$\mathbf{0.8928 \pm 0.0042}$
External regression baselines			
TIDNet	thinker-invariant CNN	0.9235 ± 0.0024	$\mathbf{0.9192 \pm 0.0027}$
EEGConformer	conv-transformer	$\mathbf{0.9188 \pm 0.0026}$	0.9287 ± 0.0057
EEGNet	compact CNN	0.9350 ± 0.0054	0.9335 ± 0.0028
LaBraM	foundation-style	0.9304 ± 0.0061	0.9327 ± 0.0086
Deep4Net	deep ConvNet	0.9269 ± 0.0045	0.9260 ± 0.0044
ShallowFBCSPNet	shallow FBCSP CNN	0.9324 ± 0.0016	0.9343 ± 0.0024
ATCNet	conv/attention/TCN	0.9686 ± 0.0183	0.9666 ± 0.0151
EEGPT	foundation-style	0.9616 ± 0.0201	0.9584 ± 0.0185
Medformer	larger transformer	0.9623 ± 0.0046	0.9585 ± 0.0051

MSP-CNN (multiscale segment-statistic pooling CNN; Figure 2) explicitly accounts for temporal structure by collecting mean and max statistics from coarse temporal segments of the feature map. These segment-level summaries act as early/middle/late activation cues and are mapped to scalar RT. ETR-CNN (event-time readout CNN) turns this coarse temporal summary into a learned time-indexed readout: it preserves temporal scores over the input window and predicts RT as the expected time under learned time-bin weights. ETR-CNN large keeps the same readout while increasing feature capacity.

We compare these task-specific regressors with external EEG architectures adapted to scalar RT regression. These include classical convolutional EEG baselines, Deep4Net and ShallowFBCSPNet (Schirrmeister et al., 2017); the compact EEGNet architecture (Lawhern et al., 2018); TIDNet's thinker-invariant convolutional design (Kostas and Rudzicz, 2020); the convolutional-transformer EEGConformer (Song et al., 2023); the attention temporal convolutional ATCNet architecture (Altaheri et al., 2022); EEGPT (Wang et al., 2024a); Medformer (Wang et al., 2024b); and the LaBraM foundation-style architecture (Jiang et al., 2024). All external architectures are trained from scratch; pretrained weights are not used.

ETR-CNN large is the strongest scalar baseline, with MSP-CNN and base ETR-CNN close behind. The external EEG backbones train meaningfully but do not close the gap to these compact temporal controls, suggesting that task-specific temporal structure matters more for this RT task than generic backbone capacity alone. These results provide strong scalar controls for the next step: testing whether directly supervising an event-time distribution improves beyond scalar RT training.

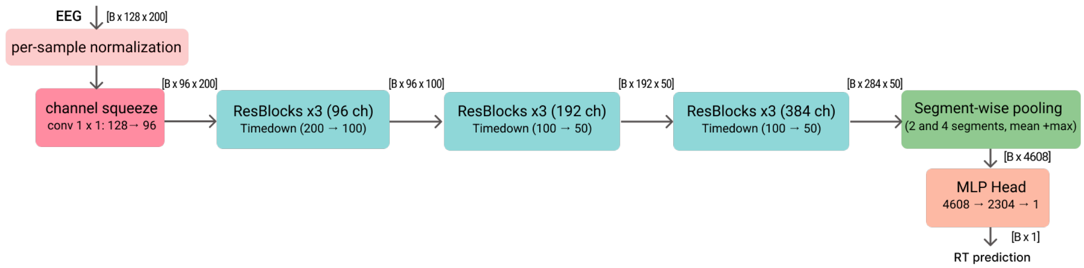

Figure 2: MSP-CNN direct RT regression baseline architecture. A lightweight 1-D temporal ConvNet maps an EEG trial window to a scalar reaction-time prediction. A learned 1×1 channel-squeeze layer (128→48) is followed by residual depthwise-separable stages with anti-aliased temporal downsampling, segment-wise pooling (2 and 4 segments; mean+max), and an MLP head.

5 Event-Time Posterior Formulation

To make response timing explicit, we reformulate RT prediction as event-time posterior modeling. Instead of predicting a scalar directly, the model produces per-time logits and a posterior distribution over response-relevant event times; scalar RT is then read out as the posterior expectation. The key distinction from temporal-readout regression is that the objective directly supervises the distributional event-time representation, rather than using a temporal head only as a scalar-regression readout.

We evaluate two ways of turning scalar RT labels into event-time supervision. The first constructs an explicit soft target distribution over time, asking the model to match a smoothed target centered at the observed RT. The second treats the observed RT statistically, as a noisy observation of an unobserved response-relevant event time, and optimizes a latent event-time likelihood. These two families correspond to soft-target objectives and likelihood-based objectives, respectively.

The core event-time experiments are designed to isolate the role of output formulation. The regression controls above test how far scalar supervision can go with task-specific temporal readouts and external EEG backbones. Here, we fix the event-time segmentation U-Net (ETS-U-Net) backbone and vary the event-time objective, so that differences in scalar accuracy, posterior geometry, and diagnostic behavior under temporal crop shifts reflect how RT is supervised over time rather than which architecture is used.

5.1 Event-Time Segmentation Architecture

ETS-U-Net maps an EEG trial window $X \in \mathbb{R}^{C \times T}$ to per-time logits $z_{0:T-1}$ (Figure 3). It follows a 1-D U-Net style encoder-decoder with skip connections for precise localization (Ronneberger et al., 2015). The encoder progressively downsamples the temporal axis to build multi-scale context, while the decoder upsamples and fuses encoder features to recover per-time outputs at the input temporal resolution (internal down/up-sampling is used only to form multi-scale features). Each scale is refined with lightweight residual blocks (ResNet-style skips) (He et al., 2016) built from depthwise-separable 1-D

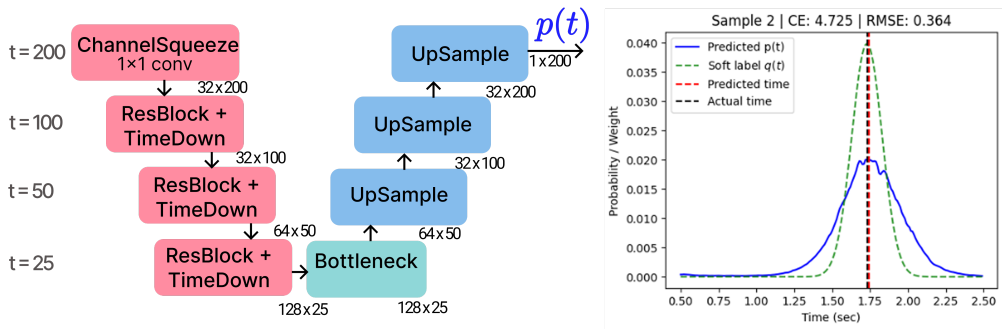

Figure 3: ETS-U-Net for RT segmentation. A lightweight 1-D U-Net maps an EEG trial window to per-time logits and an event-time distribution $p(t)$. The encoder applies a 1×1 channel-squeeze layer followed by residual blocks with temporal downsampling (`TimeDown`); a bottleneck aggregates context, and the decoder upsamples with skip connections to recover time-resolved outputs. The right panel shows an example prediction $p(t)$ and Gaussian soft target $q(t)$, together with the resulting predicted vs. observed response time.

convolutions for computational efficiency (Howard et al., 2017; Chollet, 2017). To mitigate aliasing under temporal decimation, downsampling applies a fixed depthwise smoothing filter followed by strided averaging ("blur-pool" style) (Zhang, 2019). A dilated bottleneck further enlarges the receptive field without additional downsampling (Yu and Koltun, 2016).

5.2 Soft-Target Event-Time Objectives

Soft-target objectives instantiate the direct-supervision route: the observed RT is converted into a smooth target distribution over the represented time grid.

Soft target distribution. For each fixed 2 s input window, the observed RT y is converted into a window-relative temporal target $y_{\mathrm{rel}} = y - 0.5$ s on the represented time grid. A smooth segmentation label is constructed using a Gaussian kernel with bandwidth σ:

$$\tilde{q}_t = \exp\left(-\frac{(g_t - y_{\mathrm{rel}})^2}{2\sigma^2}\right), \qquad q_t = \frac{\tilde{q}_t}{\sum_{k=0}^{T-1} \tilde{q}_k}, \qquad \sum_{t=0}^{T-1} q_t = 1. \tag{1}$$

The Gaussian label avoids a degenerate single-bin target while keeping probability mass concentrated near the observed behavioral latency.

Model output and losses. Given logits z_t for $t = 0, \ldots, T-1$, the predicted event-time distribution is obtained via temperature-scaled softmax,

$$p_t(\tau) = \frac{e^{z_t/\tau}}{\sum_{k=0}^{T-1} e^{z_k/\tau}}. \tag{2}$$

The soft-label segmentation family is written as a weighted objective:

$$\begin{aligned}\mathscr{L} = {} & \lambda_{\text{time}}\,\mathrm{RMSE}\big(\hat{t}_{\text{rel}}(\tau), y_{\text{rel}}\big) + \lambda_{\text{CE}}\,\mathrm{CE}\big(q, p(\tau)\big) \\ & + \lambda_{W1}\,W_1\big(\mathrm{CDF}(q), \mathrm{CDF}(p(\tau))\big) + \lambda_{\text{KL}}\,D_{\text{KL}}\big(q \| p(\tau)\big) + \lambda_H\,H\big(p(\tau)\big),\end{aligned} \tag{3}$$

where

$$\mathrm{CE}(q,p) = -\sum_{t=0}^{T-1} q_t \log p_t,\; D_{\text{KL}}(q\|p) = \sum_{t=0}^{T-1} q_t \log \frac{q_t}{p_t},\; H(p) = -\sum_{t=0}^{T-1} p_t \log p_t.$$

This expression defines an ablation family rather than a single prescribed loss. CE-only supervision sets the scalar time term to zero and asks the model to match a Gaussian soft event-time label. The time-only ablation removes distributional matching while keeping the same readout, and the Wasserstein-only ablation tests whether a CDF-based geometry loss is sufficient on its own. A CE+time hybrid is available by enabling both CE and soft-argmax time terms, but is not a separate row in the five-seed comparison. Entropy, KL, focal, and related terms serve as posterior-geometry controls when enabled.

5.3 Likelihood-Based Event-Time Objectives

As a probabilistic alternative to soft-target supervision, likelihood-based objectives model the observed scalar RT as a noisy realization of a latent event time. Given the posterior over latent event bins, the marginal likelihood of the observed RT is

$$p(y \mid X) = \sum_{t=0}^{T-1} p_t(X)\,\mathscr{N}\left(y; t_0 + g_t, \sigma_y^2\right), \qquad \mathscr{L}_{\text{EventNLL}} = -\log p(y \mid X). \tag{4}$$

This objective keeps the same event-time interpretation but replaces the constructed target distribution q_t with a marginal likelihood over latent response-relevant event times.

Observation-kernel and hazard extensions. The EventNLL formulation separates the latent event-time posterior $p_t(X)$ from the observation model $K(y \mid t)$. The default model uses a single Gaussian observation kernel, but we also evaluate a two-scale Gaussian mixture,

$$K(y \mid t) = (1-\alpha)\,\mathscr{N}(y; t_0 + g_t, \sigma^2_{\text{narrow}}) + \alpha\,\mathscr{N}(y; t_0 + g_t, \sigma^2_{\text{wide}}), \tag{5}$$

which keeps most probability mass in a narrow timing error while allowing a wider tail for occasional noisy RT observations. In the completed run, $\sigma_{\text{narrow}} = 0.10$ s, $\sigma_{\text{wide}} = 0.35$ s, and $\alpha = 0.10$.

As a readout extension, we also evaluate a hazard/survival parameterization. Instead of using logits only as unconstrained softmax scores, each temporal logit defines a conditional event probability

$$h_t = \sigma(z_t/\tau), \qquad p_t^{\text{haz}} \propto h_t \prod_{k<t}(1-h_k), \tag{6}$$

with the resulting event-bin probabilities normalized over the represented window. This keeps the same event-time likelihood in Eq. 4, but changes the posterior parameterization from a categorical softmax to a discrete-time survival process. This extension tests whether the event-time formulation is tied to one softmax implementation.

5.4 Formulation Comparison and Robustness

Table 4 compares the paper-facing ETS-U-Net objective variants under the fixed backbone and release-separated protocol. Here, holdout τ-nRMSE denotes holdout nRMSE after applying the posterior-readout temperature τ selected on R9–R10; R11 labels are used only for evaluation. The CE and likelihood-family objectives form the strongest scalar-readout group after readout-temperature tuning, with holdout τ-nRMSE between 0.8745 and 0.8778. This improves over the strongest scalar regression baseline, ETR-CNN large (0.8928 $\pm$ 0.0042; Table 3), while the time-only and Wasserstein controls remain closer to the scalar-regression regime.

Table 4: Event-time objective variants with a fixed ETS-U-Net backbone. Values are mean $\pm$ sample standard deviation.

Objective	**Supervision signal**	**Valid nRMSE**	**Holdout nRMSE**	**Holdout τ-nRMSE**
Soft-target objectives				
CE	Gaussian soft label	0.8763 $\pm$ 0.0044	**0.8774 $\pm$ 0.0044**	0.8753 $\pm$ 0.0039
Likelihood-based objectives				
EventNLL	Gaussian RT likelihood	0.8769 $\pm$ 0.0030	0.8805 $\pm$ 0.0021	0.8772 $\pm$ 0.0018
Mixture EventNLL	two-scale RT likelihood	**0.8744 $\pm$ 0.0018**	0.8785 $\pm$ 0.0047	**0.8745 $\pm$ 0.0053**
Hazard EventNLL	hazard posterior + likelihood	0.8755 $\pm$ 0.0027	0.8776 $\pm$ 0.0031	0.8778 $\pm$ 0.0041
Scalar-readout and distance controls				
Time-only	posterior-mean RT loss	0.8944 $\pm$ 0.0048	0.8943 $\pm$ 0.0025	0.8917 $\pm$ 0.0046
Wasserstein	CDF match to soft label	0.8997 $\pm$ 0.0035	0.8995 $\pm$ 0.0078	0.8896 $\pm$ 0.0033

Objective ablations. The ablations separate expectation-style temporal readout from distributional supervision. The time-only model directly optimizes soft-argmax error but removes distribution matching; it remains close to the strongest scalar-regression baseline and trails the CE/EventNLL group by roughly 0.014–0.017 holdout τ-nRMSE. CE and the likelihood-family variants form a tight top group: mixture EventNLL has the best mean temperature-tuned scalar readout, but CE, Gaussian EventNLL, mixture EventNLL, and hazard EventNLL are close relative to seed variability. The hazard result shows that the likelihood-family conclusion is not tied to a categorical softmax posterior,

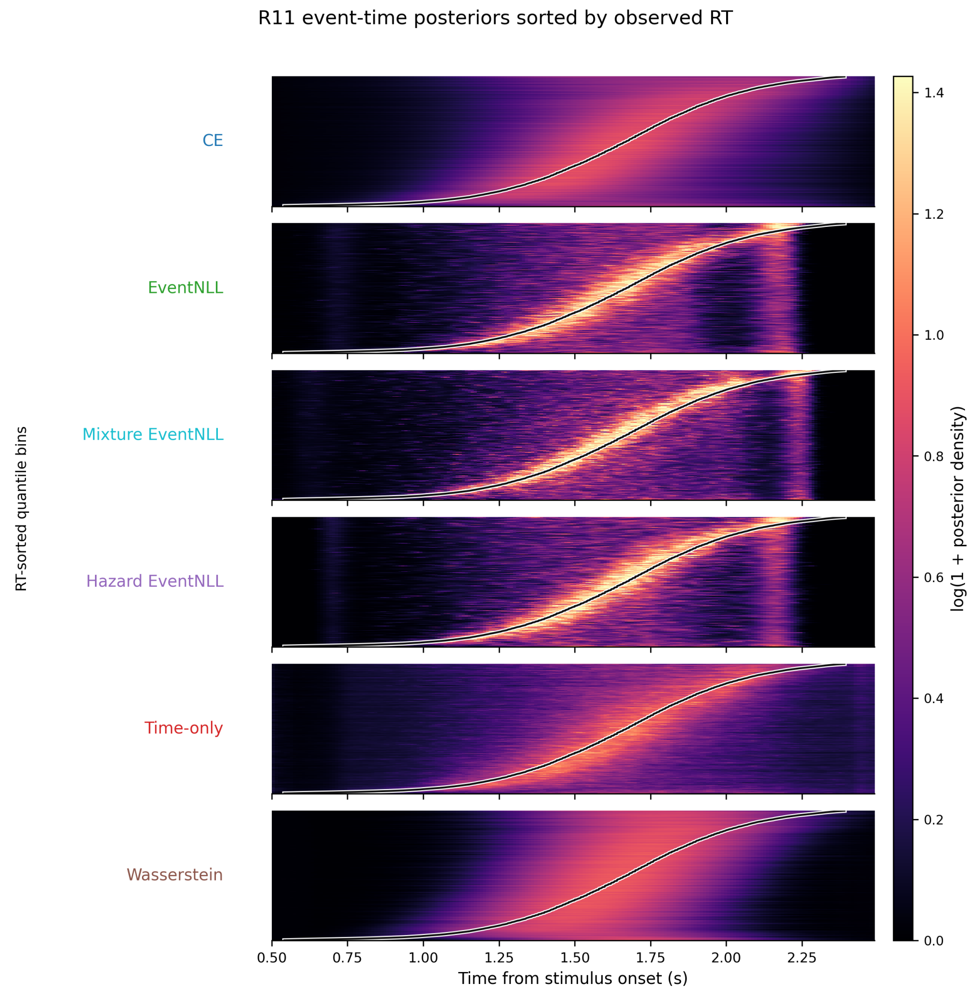

Figure 4: Trial-sorted event-time posterior maps on R11 for matched segmentation losses. Rows are quantile bins of representable R11 trials sorted by observed RT; the x-axis is time from stimulus onset; color shows log-transformed posterior density averaged within each bin. The black curve marks the mean observed RT in each bin.

while the Wasserstein control shows that an alternative distributional distance alone does not match the CE/EventNLL scalar readouts. Thus, the central gain is best attributed to distributional event-time supervision rather than to the U-Net backbone or soft-argmax readout alone.

6 Posterior Readout and Diagnostics

6.1 Scalar Readout and Temperature Tuning

Event-time models output a posterior over response-relevant time bins, whereas the benchmark evaluates scalar RT. We therefore use the posterior expectation as the scalar readout: the predicted relative time is the probability-weighted average of the temporal grid, and the absolute RT is obtained by adding the fixed window start time,

$$\hat{t}_{\mathrm{rel}}(\tau) = \sum_{t=0}^{T-1} p_t(\tau)\, g_t, \qquad \hat{t}_{\mathrm{abs}}(\tau) = t_0 + \hat{t}_{\mathrm{rel}}(\tau), \tag{7}$$

where $g_t = t\,\Delta t$ denotes the discrete time grid. This readout is aligned with squared-error evaluation and keeps event-time models comparable to scalar regression baselines under nRMSE.

Because different objectives can produce posteriors with different sharpness, we select a readout temperature on the R9–R10 development split to minimize nRMSE of the posterior-mean prediction, and apply the selected temperature unchanged to the R11 holdout set:

$$\tau_m^{\star} = \arg\min_{\tau \in \mathscr{T}_{\mathrm{cal}}} \ \mathrm{nRMSE}_{\mathrm{val}}\big(\hat{t}_{\mathrm{abs}}^{(m)}(\tau)\big). \tag{8}$$

The resulting scalar value is reported as holdout τ-nRMSE in Table 4. This temperature step is a scalar-readout protocol, not full probabilistic calibration; likelihood, coverage, and posterior concentration are evaluated separately below.

6.2 Posterior Geometry Diagnostics

Once the scalar readout has been fixed, the event-time posterior makes temporally resolved evaluation possible rather than limiting assessment to point prediction. Scalar nRMSE asks whether the posterior mean predicts RT accurately, but similar nRMSE values can hide very different temporal evidence. Posterior geometry asks where the model places probability mass, whether that mass is localized near the observed behavioral latency, whether the posterior peak agrees with the posterior mean, and whether nominal posterior intervals have meaningful empirical coverage.

We first inspect the trial-sorted posterior raster in Figure 4 before reducing the posterior to summary metrics. This dataset-level view shows how each loss distributes probability mass across the represented RT range, making posterior width, target alignment, and loss-specific confidence patterns visible. Figure 5 and Table 5 then summarize these differences quantitatively.

Table 5 summarizes posterior geometry after R9–R10 readout-temperature selection. Scalar nRMSE measures posterior-mean RT prediction, fixed-kernel EventNLL scores the full posterior under a common Gaussian observation kernel, Width80 and Mass $\pm$150 ms describe posterior concentration and target alignment, and Coverage80 and Coverage MAE summarize interval behavior. Coverage80 is the fraction of trials whose observed RT falls inside the nominal central 80% posterior interval, and Coverage MAE is the mean absolute deviation between nominal and empirical coverage across central intervals $\{50\%, 60\%, 70\%, 80\%, 90\%\}$.

These diagnostics show that losses with similar scalar error can induce different posterior semantics. CE and mixture EventNLL provide the strongest scalar readouts. EventNLL-family objectives are sharper and more target-concentrated, but they undercover nominal intervals. The soft-argmax RT-loss control has the lowest Coverage

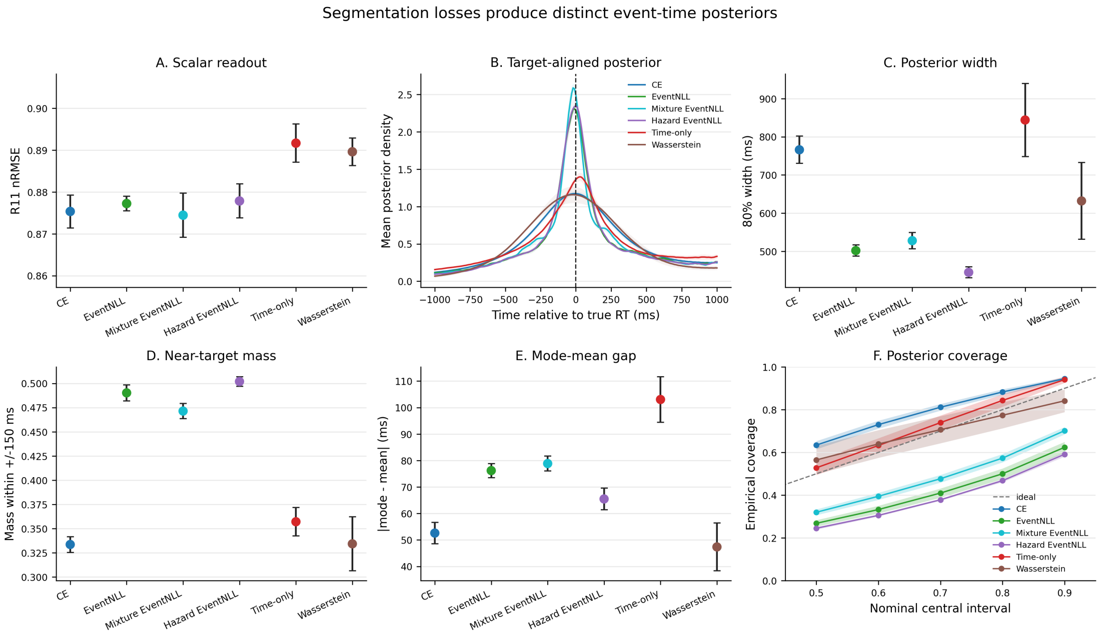

Figure 5: Posterior geometry differs across event-time losses even when scalar nRMSE is similar. Panel A reports R11 scalar readout after selecting the softmax temperature on R9–R10 to minimize nRMSE. Panel B aligns predicted posteriors to the observed RT. Panels C–E summarize posterior width, near-target mass, and mode-mean disagreement. Panel F compares empirical coverage of central posterior intervals; the diagonal is ideal interval coverage.

Table 5: Quantitative posterior geometry on R11 for matched event-time segmentation losses. Values are mean $\pm$ sample standard deviation across seeds after R9–R10 readout-temperature selection. nRMSE measures posterior-mean scalar prediction; fixed-kernel EventNLL scores the full posterior distribution under a common Gaussian observation kernel for all models ($\sigma = 0.12$ s); Width80, near-target mass, and interval coverage summarize posterior concentration and empirical reliability. For Coverage80, values closer to 0.80 indicate better nominal 80% interval coverage; for Coverage MAE, lower is better.

Objective	**nRMSE**	**Fixed NLL**	**Width80 ms**	**Mass $\pm$150 ms**	**Coverage80**	**Coverage MAE**
CE	0.875 ± 0.004	0.08 ± 0.01	766 ± 36	0.334 ± 0.008	0.883 ± 0.013	0.101 ± 0.014
EventNLL	0.877 ± 0.002	-0.06 ± 0.01	502 ± 15	0.490 ± 0.008	0.500 ± 0.025	0.273 ± 0.021
Mixture EventNLL	0.874 ± 0.005	-0.08 ± 0.01	528 ± 22	0.471 ± 0.008	0.573 ± 0.020	0.207 ± 0.017
Hazard EventNLL	0.878 ± 0.004	-0.06 ± 0.01	445 ± 14	0.502 ± 0.005	0.468 ± 0.007	0.302 ± 0.006
Soft-argmax RT loss	0.892 ± 0.005	0.11 ± 0.03	844 ± 96	0.357 ± 0.015	0.843 ± 0.029	0.040 ± 0.024
Wasserstein	0.890 ± 0.003	0.20 ± 0.04	632 ± 101	0.334 ± 0.028	0.774 ± 0.061	0.059 ± 0.021

MAE because its posterior is broad, but it is weaker as a scalar predictor and lacks distributional event-time supervision. Wasserstein is closest to nominal 80% coverage, but it has weaker scalar accuracy and lower near-target mass. Thus, posterior geometry reveals trade-offs among scalar accuracy, target concentration, distributional scoring, and interval behavior that scalar nRMSE alone would hide.

6.3 Shifted-Crop Shortcut-vs-Localization Diagnostic

Posterior geometry characterizes how probability mass is arranged inside the standard stimulus-locked window. It does not, by itself, test whether the prediction behaves like an event localizer when the temporal frame of reference changes. A fixed-window model can still exploit shortcuts: trial difficulty, subject-level response tendency, or stimulus-locked timing structure may support scalar RT prediction without requiring within-window event localization.

We therefore use shifted-crop inference as a shortcut-vs-localization diagnostic on the holdout split. For each trial, we take the same 5 s EEG cache and evaluate the trained model on 2 s crops starting at $s \in \{0.2, 0.3, \ldots, 0.8\}$ s after stimulus onset. The standard inference window corresponds to $s = 0.5$ s. For a crop starting at s, the crop-relative target is $\mathrm{RT} - s$. A model that localizes a response-relevant event within the crop should adjust its crop-relative prediction when the crop start changes; a shortcut model tied to stimulus-locked scalar timing should be less sensitive to this perturbation.

Table 6 summarizes this diagnostic together with a matched shift-jitter intervention. Shifted relative nRMSE measures crop-relative prediction accuracy across shifted crops. Sensitivity quantifies the fraction of the imposed crop shift reflected in the prediction, with 1 indicating ideal crop-relative localization and 0 indicating crop-invariant behavior. Direction is the fraction of shifted examples whose prediction moves in the expected localizer direction. Accuracy metrics are computed over crop examples in which the behavioral response remains observable within the evaluated 2 s window, whereas sensitivity and direction use the common trial subset for which the response remains inside every evaluated crop.

Table 6: Shifted-crop diagnostic and shift-jitter intervention on the 5 s holdout dataset. Values are mean $\pm$ standard deviation across seeds. Holdout τ-nRMSE uses the canonical fixed window; shifted relative nRMSE is computed on shifted-crop examples in which RT remains observable within the evaluated 2 s window; sensitivity and direction are computed on the common-inside subset.

Objective	Holdout τ-nRMSE $\downarrow$		Shifted rel. nRMSE $\downarrow$		Sensitivity $\uparrow$		Direction $\uparrow$	
	Fixed	Jitter	Fixed	Jitter	Fixed	Jitter	Fixed	Jitter
CE	0.8753 ± 0.0039	0.8749 ± 0.0060	0.8679 ± 0.0024	0.8569 ± 0.0060	0.5810 ± 0.0351	0.5831 ± 0.0406	0.7780 ± 0.0041	0.7924 ± 0.0056
Mixture EventNLL	0.8745 ± 0.0053	0.8734 ± 0.0033	0.8663 ± 0.0037	0.8576 ± 0.0041	0.6021 ± 0.0484	0.6249 ± 0.0313	0.7730 ± 0.0062	0.7925 ± 0.0076
EventNLL	0.8772 ± 0.0018	0.8771 ± 0.0040	0.8685 ± 0.0023	0.8593 ± 0.0028	0.5842 ± 0.0287	0.6174 ± 0.0222	0.7739 ± 0.0059	0.7937 ± 0.0068
Hazard EventNLL	0.8778 ± 0.0041	0.8806 ± 0.0034	0.8692 ± 0.0028	0.8609 ± 0.0045	0.5618 ± 0.0376	0.5891 ± 0.0259	0.7694 ± 0.0090	0.7925 ± 0.0074
Soft-argmax RT loss	0.8917 ± 0.0046	0.8836 ± 0.0035	0.8857 ± 0.0069	0.8613 ± 0.0032	0.5381 ± 0.0586	0.5775 ± 0.0362	0.7592 ± 0.0183	0.7951 ± 0.0052
Wasserstein	0.8896 ± 0.0033	0.8922 ± 0.0066	0.8932 ± 0.0033	0.8774 ± 0.0099	0.6684 ± 0.0711	0.6852 ± 0.0416	0.7830 ± 0.0197	0.8026 ± 0.0118

We then trained matched shift-jitter variants in which the 2 s training window was sampled from the same crop-start range and the target was expressed relative to the sampled window. The intervention does not produce a broad ordinary-holdout gain under the canonical support: CE, mixture EventNLL, and Gaussian EventNLL remain essentially unchanged in holdout τ-nRMSE. Its main effect is instead on shifted-crop behavior. Shifted relative nRMSE improves for every objective, and direction consistently increases, indicating that predictions more often move in the expected localizer direction. Sensitivity also improves modestly, but remains well below the ideal value of 1.

Across objectives, Wasserstein is the most localizer-like by sensitivity and direction, but this comes with weaker scalar and shifted-crop accuracy. This reinforces the broader trade-off exposed by the posterior analyses: objectives that encourage stronger crop-relative movement need not be the best point predictors. Overall, shift-jitter reduces fixed-window shortcut behavior without turning these models into fully shift-equivariant event localizers.

7 Discussion

7.1 Event-Time Supervision and Objective Geometry

The results support treating trial-wise RT prediction as event-time posterior modeling rather than as ordinary scalar regression. The label is a behavioral latency within a post-stimulus interval, and models benefit when that temporal structure is explicit in the output space and in the loss. CE and the EventNLL-family objectives form the strongest scalar-readout group, whereas the soft-argmax RT-loss control is weaker despite using the same posterior-mean readout. This supports the interpretation that the gain comes from distributional event-time supervision, not merely from replacing a scalar head with a differentiable expectation.

The objective comparison also shows that event-time losses do not all impose the same computational semantics. CE-style soft targets provide strong scalar readouts and broad posterior support. EventNLL reaches a similar scalar-error regime through a latent-variable likelihood, marginalizing an unobserved response-relevant event time under an observation model for RT. Wasserstein behaves differently: it is more localizer-like by shifted-crop sensitivity and direction, but weaker as a point predictor. Thus, the choice of objective controls a trade-off among scalar accuracy, posterior concentration, interval behavior, and crop-relative movement.

The wrapped external baselines show that standardized EEG architectures can train under the same fixed-normalization protocol, but they do not close the gap to compact task-specific models or event-time segmentation. Foundation-style architectures are evaluated here as architectures trained from scratch, not as pretrained models. The comparison is therefore not a test of pretraining; it shows that, under this benchmark protocol, formulation and objective design are strong levers even with lightweight supervised models.

7.2 Posterior Diagnostics and Shifted-Crop Behavior

The event-time posterior contains more information than its expectation. Posterior width, target-aligned mass, mode-mean disagreement, CRPS, fixed-kernel EventNLL, and interval coverage expose temporal output behavior that scalar nRMSE discards. This is important because similar point errors can correspond to different posterior semantics: broad conservative distributions, sharper target-concentrated distributions, or better interval coverage with weaker scalar prediction.

Temperature tuning should therefore be interpreted narrowly. In this study, temperature is a scalar-readout protocol for comparing posterior-mean predictions under nRMSE. It is not the main methodological claim and does not replace distributional diagnostics. Likelihood, coverage, posterior concentration, and shifted-crop behavior are evaluated separately because they answer different questions about the learned event-time posterior.

The shifted-crop diagnostic extends this posterior analysis from shape to behavior under temporal perturbation. If fixed-window training permits stimulus-locked shortcuts, then crop shifts should reveal whether predictions move in a crop-relative, localizer-like direction. Shift-jitter training targets this failure mode directly by varying the crop start during training and expressing the target relative to the sampled crop. Its main effect is not a broad improvement in ordinary fixed-window holdout nRMSE, but improved shifted-crop robustness and more frequent movement in the expected localizer direction. Sensitivity remains well below the ideal value of 1, so shift-jitter reduces fixed-window shortcut behavior without solving fully shift-equivariant event localization.

7.3 Scope and Limitations

The main limitation is that all claims are tied to the HBN-EEG contrast change detection task and the release-separated protocol used here. R11 is a stronger local holdout than reused validation performance, but it is still drawn from the same broader dataset family and preprocessing pipeline. Broader claims about EEG event-time modeling require replication across tasks, acquisition settings, montages, and latency targets.

The shifted-crop results should also be interpreted as partial localization evidence rather than solved event detection. Even after shift-jitter training, sensitivity remains far from the ideal crop-relative localizer. Stronger localization claims will require training objectives, architectures, or augmentations designed explicitly for temporal equivariance. The five-seed robustness checks support the main scalar-versus-event-time comparison, but they cover selected core recipes rather than every kernel, hazard, calibration, and architecture variant. The foundation-style comparisons also do not use pretrained weights or montage-specialized setup changes; they are controlled architecture baselines rather than a full evaluation of EEG foundation models.

Taken together, these results support event-time posterior modeling as a computational bridge between behavioral latency prediction and event-centered EEG analysis. The framework improves scalar RT decoding, exposes posterior structure hidden by nRMSE, and enables shortcut-vs-localization diagnostics under temporal perturbation. At the same time, the shifted-crop results make the remaining gap explicit: current fixed-window models become more localizer-like, especially under shift-jitter training, but they do not yet implement full crop-relative event localization.

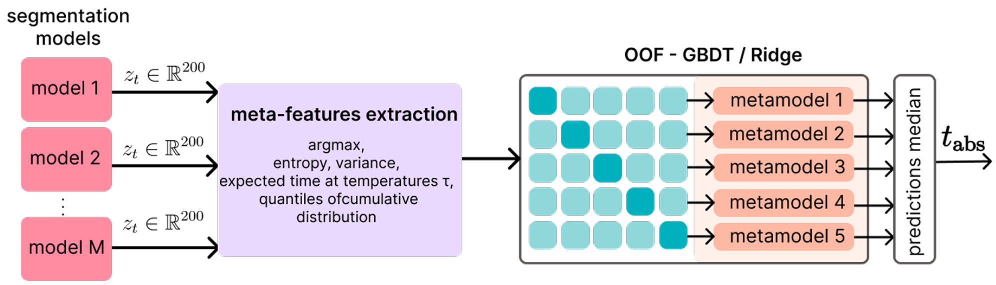

Figure 6: OOF stacking for RT prediction from time-distribution outputs. Per-model logits/distributions are converted into meta-features (e.g., peak time, entropy/variance, temperature-conditioned expected times, and CDF quantiles) and combined by a second-stage stacker (Ridge/GBDT). Predictions from fold-trained stackers are aggregated at inference (median) to produce the final RT estimate.

8 Conclusion

We reformulated EEG-based reaction-time prediction as event-time distribution modeling and evaluated it under a release-separated HBN-EEG protocol. The results indicate that supervising an event-time posterior improves RT prediction beyond scalar regression and beyond a temporal-readout head alone. Loss ablations and five-seed robustness checks show that distributional event-time supervision matters, EventNLL provides a principled latent-variable likelihood alternative, and posterior-geometry plus shifted-crop analyses reveal output behavior that scalar nRMSE alone cannot capture.

A Distribution-Aware Stacking

Segmentation-style outputs enable richer ensembling than scalar averaging, but combination becomes sensitive to model confidence: naive averaging can overweight overconfident models and underweight accurate but underconfident ones. We therefore define an auxiliary second-stage stacker on distributional meta-features extracted from the time-distribution outputs.

In a release-separated application, base segmentation models are fit on R1-R8 before the stacker is trained. Meta-model inputs are then computed on the R9-R10 development partition using subject-disjoint folds, so that the stacker never receives in-fold predictions for the same subject. Logits and temperature-scaled event-time probabilities are converted to meta-features such as peak time, entropy, variance, expected time, and CDF quantiles. This stacker is treated as a post-hoc ensemble layer rather than as part of the single-model event-time comparison.

Meta-features from logits and temperature-scaled probabilities. Meta-features are extracted per model from (i) raw logits and (ii) temperature-scaled probabilities computed for multiple temperatures $\tau \in \mathscr{T}_{\text{meta}}$. From logits, a hard-peak estimate and confidence proxies are computed:

$$t^\star = \arg\max_t z_t, \qquad \hat{t}_{\text{hard}} = t_0 + g_{t^\star}, \qquad z_{\max} = \max_t z_t, \qquad m_{\text{top2}} = z_{\max} - \max_{t \neq t^\star} z_t. \quad (9)$$

For each $\tau \in \mathscr{T}_{\text{meta}}$, probabilities are formed as $p(\tau) = \text{softmax}(\mathbf{z}/\tau)$ (Eq. 2, Sec. 5), where $\mathbf{z} = (z_0, \ldots, z_{T-1})$ denotes the per-time logit vector for a given model and window. Distributional summaries are computed, including the soft-argmax time (Eq. 7, Sec. 5), entropy $H(p(\tau))$, peak probability $p_{\max}(\tau)$, time variance $\text{Var}_{p(\tau)}[g]$, and CDF time-quantiles (10/50/90%). In our implementation, $\mathscr{T}_{\text{meta}} = \{0.7, 1.0, 1.3\}$.

Meta-learner and inference. We use linear Ridge regression as a strong baseline meta-learner, and gradient boosting to capture non-linear reliability patterns across models and temperatures. The meta-learner is trained using subject-disjoint cross-validation: each subject is assigned to exactly one fold (Figure 6). For fold k, the stacker is fit on the other folds and predicts RT for the held-out fold. The leakage-controlled diagnostic is the out-of-fold nRMSE from these held-out predictions. Folds are stratified to match the per-subject mean RT distribution across folds. For held-out inference, the same feature construction is applied to unseen logits; predictions from the fold-trained stackers are aggregated to obtain the final RT estimate. Manual blending serves as a reference, but it cannot adapt weights to sample-specific confidence profiles.

B Original Hidden CodaBench Benchmark Results

We report the original hidden-release CodaBench evaluation as historical challenge context. These scores provide continuity with the original benchmark setting, while the main empirical conclusions in this manuscript are based on the labeled R11 release with subject-bootstrap confidence intervals.

Table 7: Original hidden-release CodaBench RT results from the earlier submission-oriented pipeline. These results are reported for historical comparison with the challenge setting.

Method	Validation nRMSE ↓	Hidden nRMSE ↓
Best temperature-tuned single model	0.898757	0.92334
Manual blending	0.899979	0.91840
Ridge stacking	0.890430	0.91539
Gradient boosting stacking	**0.883365**	**0.91394**

C Unwrapped External Backbone Sanity Checks

Unwrapped external architectures are useful for diagnosing scale sensitivity. Because our own models use per-window standardization, the primary comparison uses wrapped

Table 8: Unwrapped external-backbone sanity checks. “Wrapped” means that the model receives the same per-window channel standardization used by our compact baselines. These checks diagnose scale sensitivity; the wrapped variants are the fair external comparisons used in the main results.

Architecture	Unwrapped valid	Unwrapped R11	Wrapped valid	Wrapped R11	Takeaway
EEGNet	1.000000	1.002046 [0.999718, 1.007253]	0.957877	0.965041 [0.954684, 0.977111]	Standardization makes the compact EEG baseline train
EEGConformer	1.000000	1.000413 [0.999906, 1.003179]	0.957968	0.962802 [0.951634, 0.975382]	Transformer-style baseline is scale-sensitive without wrapping
LaBraM	1.000000	1.000711 [0.998638, 1.004512]	0.958254	0.965382 [0.952576, 0.979330]	From-scratch foundation-style architecture needs the fixed normalization

Table 9: Additional EventNLL kernel and hazard robustness checks. “Cal.” denotes post-hoc temperature selection on R9–R10; the heteroscedastic run is reported at its base readout because it was not part of the calibrated kernel table.

Variant	Change tested	τ	Valid nRMSE ↓	R11 nRMSE ↓	Takeaway
Gaussian EventNLL, cal.	fixed Gaussian observation kernel	0.70	0.923619	0.940947 [0.923160, 0.960322]	Core latent event-time likelihood reference
Gaussian mixture EventNLL, cal.	narrow/wide Gaussian observation mixture	0.75	0.921676	0.939267 [0.921864, 0.958291]	Strongest calibrated EventNLL-family scalar readout
Laplace EventNLL, cal.	Laplace observation kernel	0.75	0.926636	0.940696 [0.923718, 0.959285]	Robust kernel did not improve over cleaner Gaussian/mixture variants
Student-t EventNLL, cal.	heavy-tailed Student-t observation kernel	0.80	0.923259	0.941127 [0.923382, 0.960687]	Calibration recovers validation gap, but R11 remains tied or weaker
Heteroscedastic EventNLL	trial-wise learned observation scale $\sigma(x)$	0.65	0.923056	0.942343 [0.923630, 0.962703]	Learned scale did not improve R11 enough for the core table
Hazard EventNLL	hazard posterior with continuous Gaussian EventNLL	0.65	0.924318	0.937332 [0.920594, 0.955704]	Strong alternative posterior parameterization
Hazard discrete NLL	exact-bin discrete survival likelihood	0.65	0.926288	0.946823 [0.928204, 0.967220]	Negative control: exact-bin survival loss is weaker than continuous EventNLL

external models. Table 8 reports matched sanity checks for representative off-the-shelf architectures with and without the fixed per-window standardization wrapper. The unwrapped rows often collapse toward target-standard-deviation behavior (nRMSE near 1), whereas the wrapped variants train meaningfully under the same release-separated protocol used in the main tables.

D Additional Protocol Calibration Runs

Protocol calibration evaluated optimizer choice, learning rate, soft-label width, batch size, and augmentation strength before fixing the training recipe used in the release-separated experiments. These diagnostics informed the protocol but are separated from the R11 results used for the main claims.

E Additional EventNLL Kernel and Hazard Robustness Checks

The main text keeps the event-time segmentation table focused on the core CE, EventNLL, and control objectives, plus two compact EventNLL-family extensions. Table 9 reports the broader kernel and hazard robustness checks. These runs use the same release-separated protocol and test whether the probabilistic event-time result depends on one specific Gaussian observation kernel or one specific softmax parameterization.

Overall, these checks support two points. First, the EventNLL view is not tied to a single Gaussian-softmax implementation: the hazard EventNLL and Gaussian-mixture kernel remain competitive. Second, more robust or more flexible observation models are not automatically better; Laplace, Student-t, heteroscedastic, and exact-bin hazard variants are useful robustness probes but do not strengthen the core R11 story enough to

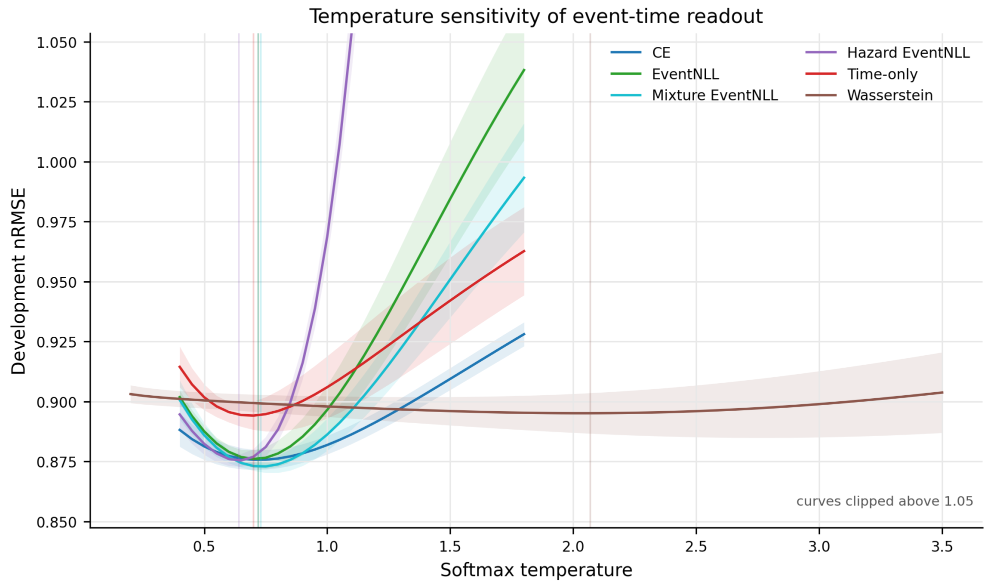

Figure 7: Temperature sensitivity on the R9-R10 development split. Vertical lines mark selected temperatures. The curves show that different losses induce different confidence scales, motivating temperature calibration as posterior-geometry control before comparing scalar readouts or using posterior features downstream.

feature in the main table.

F Additional Architecture Ablations

Architecture ablations separate formulation effects from capacity and backbone-specific effects. In particular, ETS-U-Net capacity variants, attention bottlenecks, factorized channel interactions, recurrent variants, and EEG-Inception-style segmentation provide complementary checks on whether the event-time formulation depends on a single encoder-decoder instantiation.

G Additional Posterior Geometry Examples

Table 5 reports the quantitative posterior metrics behind Figure 5, and Figure 4 shows the corresponding trial-sorted posterior maps in the main text. Figure 7 shows the development-split temperature curves used to choose post-hoc readout temperatures.

References

Altaheri, H., Muhammad, G., and Alsulaiman, M. (2022). Physics-informed attention temporal convolutional network for EEG-based motor imagery classification. *IEEE Transactions on Industrial Informatics*.

Aristimunha, B., Truong, D., Guetschel, P., Shirazi, S. Y., Guyon, I., Franco, A. R., Milham, M. P., Dotan, A., Makeig, S., Gramfort, A., King, J.-R., Corsi, M.-C., Valdés-Sosa, P. A., Majumdar, A., Evans, A., Sejnowski, T. J., Shriki, O., Chevallier, S., and Delorme, A. (2025). Eeg foundation challenge: From cross-task to cross-subject eeg decoding.

Barachant, A., Bonnet, S., Congedo, M., and Jutten, C. (2012). Multiclass brain–computer interface classification by Riemannian geometry. *IEEE Transactions on Biomedical Engineering*, 59(4):920–928.

Bigdely-Shamlo, N., Cockfield, J., Makeig, S., Rognon, T., La Valle, C., Miyakoshi, M., and Robbins, K. A. (2016). Hierarchical event descriptors (HED): Semi-structured tagging for real-world events in large-scale EEG. *Frontiers in Neuroinformatics*, 10:42.

Chambon, S., Thorey, V., Arnal, P. J., Mignot, E., and Gramfort, A. (2019). DOSED: A deep learning approach to detect multiple sleep micro-events in EEG signal. *Journal of Neuroscience Methods*, 321:64–78.

Chapfuwa, P., Tao, C., Li, C., Khan, I., Chandross, K. J., Pencina, M. J., Carin, L., and Henao, R. (2023). Calibration and uncertainty in neural time-to-event modeling. *IEEE Transactions on Neural Networks and Learning Systems*, 34(4):1666–1680.

Chen, Y., Ren, K., Song, K., Wang, Y., Wang, Y., Li, D., and Qiu, L. (2024). Eegformer: Towards transferable and interpretable large-scale eeg foundation model.

Chollet, F. (2017). Xception: Deep learning with depthwise separable convolutions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Chowdhury, M. S. N., Dutta, A., Robison, M. K., Blais, C., Brewer, G., and Bliss, D. W. (2020). Deep neural network for visual stimulus-based reaction time estimation using the periodogram of single-trial EEG. *Sensors*, 20(21):6090.

Coles, M. G. H. (1989). Modern mind-brain reading: Psychophysiology, physiology, and cognition. *Psychophysiology*, 26(3):251–269.

Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society: Series B (Methodological)*, 34(2):187–202.

EEG Challenge (2025). Eeg challenge (2025): Neurips eeg foundation challenge. `https://eeg2025.github.io/`. Accessed 2026-02-04.

Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., Marchand, M., and Lempitsky, V. (2016). Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17:1–35. arXiv:1505.07818.

Gehring, W. J., Goss, B., Coles, M. G. H., Meyer, D. E., and Donchin, E. (1993). A neural system for error detection and compensation. *Psychological Science*, 4(6):385–390.

Geng, X. (2016). Label distribution learning. *IEEE Transactions on Knowledge and Data Engineering*, 28(7):1734–1748.

Haider, H., Hoehn, B., Davis, S., and Greiner, R. (2020). Effective ways to build and evaluate individual survival distributions. *Journal of Machine Learning Research*, 21(85):1–63.

He, H. and Wu, D. (2020). Transfer learning for brain–computer interfaces: A Euclidean space data alignment approach. *IEEE Transactions on Biomedical Engineering*, 67(2):399–410.

He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Holroyd, C. B. and Coles, M. G. H. (2002). The neural basis of human error processing: Reinforcement learning, dopamine, and the error-related negativity. *Psychological Review*, 109(4):679–709.

Howard, A. G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M., and Adam, H. (2017). Mobilenets: Efficient convolutional neural networks for mobile vision applications. *CoRR*, abs/1704.04861.

Jayaram, V., Alamgir, M., Altun, Y., Schölkopf, B., and Grosse-Wentrup, M. (2016). Transfer learning in brain–computer interfaces. *IEEE Computational Intelligence Magazine*, 11(1):20–31.

Jiang, W., Zhao, L., and liang Lu, B. (2024). Large brain model for learning generic representations with tremendous EEG data in BCI. In *The Twelfth International Conference on Learning Representations*.

Jung, T.-P., Makeig, S., Westerfield, M., Townsend, J., Courchesne, E., and Sejnowski, T. J. (1998). Analyzing and visualizing single-trial event-related potentials. In *Advances in Neural Information Processing Systems*, volume 11, pages 118–124.

Kelly, S. P. and O'Connell, R. G. (2013). Internal and external influences on the rate of sensory evidence accumulation in the human brain. *Journal of Neuroscience*, 33(50):19434–19441.

Kornhuber, H. H. and Deecke, L. (1965). Hirnpotentialaenderungen bei willkuerbewegungen und passiven bewegungen des menschen: Bereitschaftspotential und reafferente potentiale. *Pfluegers Archiv fuer die gesamte Physiologie des Menschen und der Tiere*, 284(1):1–17.

Kostas, D., Aroca-Ouellette, S., and Rudzicz, F. (2021). BENDR: Using transformers and a contrastive self-supervised learning task to learn from massive amounts of EEG data. *Frontiers in Human Neuroscience*, 15:653659.

Kostas, D. and Rudzicz, F. (2020). Thinker invariance: enabling deep neural networks for BCI across more people. *Journal of Neural Engineering*, 17:056008.

Lawhern, V. J., Solon, A. J., Waytowich, N. R., et al. (2018). Eegnet: a compact convolutional neural network for eeg-based brain–computer interfaces. *Journal of Neural Engineering*.

Lee, C., Zame, W. R., Yoon, J., and van der Schaar, M. (2018). DeepHit: A deep learning approach to survival analysis with competing risks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, pages 2314–2321.

O'Connell, R. G., Dockree, P. M., and Kelly, S. P. (2012). A supramodal accumulation-to-bound signal that determines perceptual decisions in humans. *Nature Neuroscience*, 15(12):1729–1735.

Ouahidi, Y. E., Lys, J., Thölke, P., Farrugia, N., Pasdeloup, B., Gripon, V., Jerbi, K., and Lioi, G. (2025). Reve: A foundation model for eeg – adapting to any setup with large-scale pretraining on 25,000 subjects.

Ouyang, G., Herzmann, G., Zhou, C., and Sommer, W. (2011). Residue iteration decomposition (RIDE): A new method to separate ERP components on the basis of latency variability in single trials. *Psychophysiology*, 48(12):1631–1647.

Ouyang, G., Hildebrandt, A., Sommer, W., and Zhou, C. (2017). Exploiting the intra-subject latency variability from single-trial event-related potentials in the P3 time range: A review and comparative evaluation of methods. *Neuroscience and Biobehavioral Reviews*, 75:1–21.

Ratcliff, R. and McKoon, G. (2008). The diffusion decision model: Theory and data for two-choice decision tasks. *Neural Computation*, 20(4):873–922.

Ronneberger, O., Fischer, P., and Brox, T. (2015). U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, volume 9351 of *Lecture Notes in Computer Science*, pages 234–241. Springer.

Schirrmeister, R. T., Springenberg, J. T., Fiederer, L. D. J., Glasstetter, M., Eggensperger, K., Tangermann, M., Hutter, F., Burgard, W., and Ball, T. (2017). Deep learning with convolutional neural networks for eeg decoding and visualization. *Human Brain Mapping*.

Shibasaki, H. and Hallett, M. (2006). What is the Bereitschaftspotential? *Clinical Neurophysiology*, 117(11):2341–2356.

Shirazi, S. Y., Franco, A., Hoffmann, M. S., Esper, N., Truong, D., Delorme, A., Milham, M., and Makeig, S. (2024). Hbn-eeg: The fair implementation of the healthy brain network (hbn) electroencephalography dataset. *bioRxiv*.

Song, Y., Zheng, Q., Liu, B., and Gao, X. (2023). Eeg conformer: Convolutional transformer for eeg decoding and visualization. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 31:710–719.

Sun, B. and Saenko, K. (2016). Deep CORAL: Correlation alignment for deep domain adaptation. In *Proceedings of the European Conference on Computer Vision (ECCV)*.

Tapia-Rivas, N. I., Estevez, P. A., and Cortes-Briones, J. A. (2024). A robust deep learning detector for sleep spindles and K-complexes: Towards population norms. *Scientific Reports*, 14:263.

Wang, G., Liu, W., He, Y., Xu, C., Ma, L., and Li, H. (2024a). Eegpt: Pretrained transformer for universal and reliable representation of eeg signals. In Globerson, A., Mackey, L., Belgrave, D., Fan, A., Paquet, U., Tomczak, J., and Zhang, C., editors, *Advances in Neural Information Processing Systems*, volume 37, pages 39249–39280. Curran Associates, Inc.

Wang, J., Zhao, S., Luo, Z., Zhou, Y., Li, S., and Pan, G. (2025). Eegmamba: An eeg foundation model with mamba. *Neural Networks*, 192:107816.

Wang, Y., Huang, N., Li, T., Yan, Y., and Zhang, X. (2024b). Medformer: A multi-granularity patching transformer for medical time-series classification. In Globerson, A., Mackey, L., Belgrave, D., Fan, A., Paquet, U., Tomczak, J., and Zhang, C., editors, *Advances in Neural Information Processing Systems*, volume 37, pages 36314–36341.

Wu, D., Lance, B. J., Lawhern, V. J., Gordon, S., Jung, T.-P., and Lin, C.-T. (2017). EEG-based user reaction time estimation using riemannian geometry features. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 25(11):2157–2168.

Yu, F. and Koltun, V. (2016). Multi-scale context aggregation by dilated convolutions. In *International Conference on Learning Representations (ICLR)*.

Yue, T., Gao, X., Xue, S., Tang, Y., Guo, L., Jiang, J., and Liu, J. (2025). Braingpt: Unleashing the potential of eeg generalist foundation model by autoregressive pre-training.

Zanini, P., Congedo, M., Jutten, C., Said, S., and Berthoumieu, Y. (2018). Transfer learning: A Riemannian geometry framework with applications to brain–computer interfaces. *IEEE Transactions on Biomedical Engineering*, 65(5):1107–1116.

Zhang, R. (2019). Making convolutional networks shift-invariant again. In Chaudhuri, K. and Salakhutdinov, R., editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 7324–7334. PMLR.